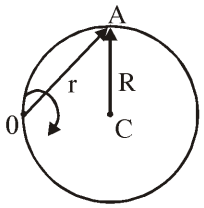
SECTION-I

Single Correct Answer Type **8 Q. [3 M (–1)]**

1. A particle A moves along a circle of radius R=50 cm so that its radius vector r relative to the point O (figure) rotates with the constant angular velocity ω=0.40 rad/s. Then modulus of the velocity of the particle, and the modulus of its total acceleration will be

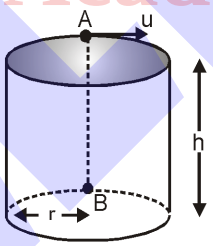


(A) v= 0.4 m/s, a = 0.4 m/s^2 (B) v = 0.32 m/s, a = 0.32 m/s^2

(C) v = 0.32 m/s, a = 0.4 m/s^2 (D) v = 0.4 m/s, a = 0.32 m/s^2

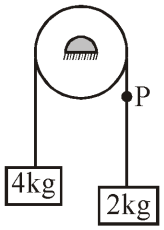
2. A hollow vertical cylinder of radius r and height h has a smooth internal surface. A small particle is placed in contact with the inner side of the upper rim, at point A, and given a horizontal speed u, tangential to the rim. It leaves the lower rim at point B, vertically below A. If n is an integer then

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(A) $\frac{u}{2\pi r}\sqrt{\frac{2h}{g}} = n$ (B) $\frac{h}{2\pi r} = n$ (C) $\frac{2\pi r}{h} = n$ (D) $\frac{u}{\sqrt{2gh}} = n$

3. Point P is on a massless thread in an ideal pulley arrangement as shown.



Statement 1 : As point P moves from right side to left side of pulley, the magnitude of it's acceleration remains same throughout.

and

Statement 2 : The tension in massless thread remains uniform in magnitude.

(A) Statement-1 is true, statement -2 is true and statement-2 is correct explanation for statement-1.

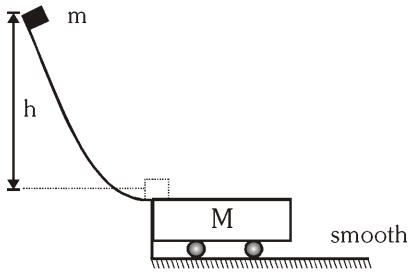
(B) Statement-1 is true, statement -2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement -2 is false.

(D) Statement-1 is false, statement -2 is true.

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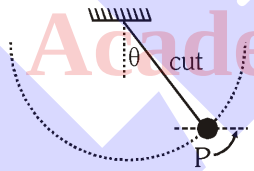
4. A carriage of mass M and length ℓ is joined to the end of a slope as shown in the figure. A block of mass m is released from the slope from height h. It slides till end of the carriage. The friction between the mass and the slope and also friction between carriage and horizontal floor is negligible. Coefficient of friction between block and carriage is μ. Find h in the given terms.



(A) $\mu\left(1+\frac{M}{m}\right)\ell$ (B) $2\mu\left(1+\frac{M}{m}\right)\ell$ (C) $\mu\left(2+\frac{m}{M}\right)\ell$ (D) $\mu\left(1+\frac{m}{M}\right)\ell$

5. Consider a pendulum, consisting of a massless string with a mass on the end. The mass is held with the string horizontal, and then released. The mass swings down, and then on its way back up, the string is cut at point P when it makes an angle of θ with the vertical. What should θ be, so that the mass travels the largest horizontal distance from P by the time it returns to the height it had when the string was cut?

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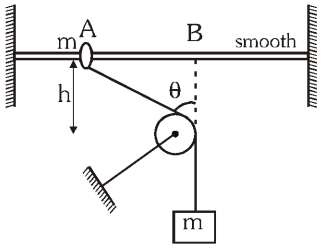


(A) $\tan^{-1}\sqrt{3}$ (B) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (C) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (D) $\tan^{-1}(1)$

6. A uniform chain has a mass M and length L. It is placed on a frictionless table with length ℓ_0 hanging over the edge. The chain begins to slide down. The speed V with which the chain slides away from the edge is given by

(A) $V=\sqrt{\frac{g\ell_0}{L}\left(L+\ell_0\right)}$ (B) $V=\sqrt{\frac{g\ell_0}{L}\left(L-\ell_0\right)}$ (C) $V=\sqrt{\frac{g}{L}\left(L^2-\ell_0^2\right)}$ (D) $V=\sqrt{2g\left(L-\ell_0\right)}$

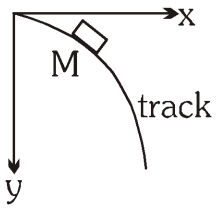
7. A ring of mass m slide from rest on the smooth rod as shown in the figure, due to the block of mass m. Pully and string are massless. Then find the speed of ring when the string become straight. (Given $\theta = 60°$)



(A) $4\sqrt{gh}$ (B) $\sqrt{2gh}$ (C) $2\sqrt{2gh}$ (D) $8\sqrt{gh}$

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8. A man M slides down a curved frictionless track, starting from rest. The curve obeys the equation $y = \frac{x^2}{2}$. The tangential acceleration of man is:



(A) g (B) $\frac{gx}{\sqrt{x^2+4}}$ (C) $\frac{g}{2}$ (D) $\frac{gx}{\sqrt{x^2+1}}$

Multiple Correct Answer Type 3 Q. [4 M (–1)]

9. An ant travels along a long rod with a constant velocity $\vec{u}$ relative to the rod starting from the origin. The rod is kept initially along the positive x-axis. At t = 0, the rod also starts rotating with an angular velocity ω (anticlockwise) in x-y plane about origin. Then

(A) the position of the ant at any time t is $\vec{r} = ut[\cos\omega t\hat{i} + \sin\omega t\hat{j}]$.

(B) the speed of the ant at any time t is $u\sqrt{1+\omega^2t^2}$.

(C) the magnitude of the tangential acceleration of the ant at any time t is $\frac{\omega^2 tu}{\sqrt{1+\omega^2t^2}}$.

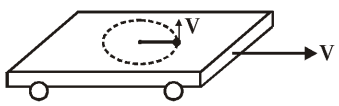
(D) the speed of the ant at any time t is $\sqrt{1+2\omega^2t^2u}$.

10. Which of the following is/are conservative force(s)?

(A) $\vec{F} = 2r^3\hat{r}$ (B) $\vec{F} = -\frac{5}{r}\hat{r}$ (C) $\vec{F} = \frac{3(x\,i + y\,\hat{j})}{(x^2+y^2)^{3/2}}$ (D) $\vec{F} = \frac{3(y\,i + x\,\hat{j})}{(x^2+y^2)^{3/2}}$

11. On a train moving along east with a constant speed v, a boy revolves a bob with string of length ℓ on smooth surface of a train, with equal constant speed v relative to train. Mark the correct option(s).

(A) Maximum speed of bob is 2 v in ground frame.



(B) Tension in string connecting bob is $\frac{4mv^2}{\ell}$ at an instant.

(C) Tension in string is $\frac{mv^2}{\ell}$ at all the moments.

(D) Minimum speed of bob is zero in ground frame.

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Linked Comprehension Type (1 Para × 2Q, 1 Para × 3 Q.) [3 M(-1)]

(Single Correct Answer Type)

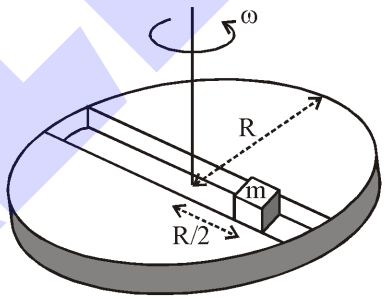
Paragraph for Question No. 12 and 13

A frame of reference that is accelerated with respect to an inertial frame of reference is called a non-inertial frame of reference. A coordinate system fixed on a circular disc rotating about a fixed axis with a constant angular velocity ω is an example of a non-inertial frame of reference. The relationship between the force $\vec{F}_{rot}$ experienced by a particle of mass m moving on the rotating disc and the force $\vec{F}_{in}$ experienced by the particle in an inertial frame of reference is

$$\vec{F}_{rot} = \vec{F}_{in} + 2m(\vec{\upsilon}_{rot} \times \vec{\omega}) + m(\vec{\omega} \times \vec{r}) \times \vec{\omega},$$

where $\vec{\upsilon}_{rot}$ is the velocity of the particle in the rotating frame of reference and $\vec{r}$ is the position vector of the particle with respect to the centre of the disc.

Now consider a smooth slot along a diameter of a disc of radius R rotating counter-clockwise with a constant angular speed ω about its vertical axis through its center. We assign a coordinate system with the origin at the centre of the disc, the x-axis along the slot, the y-axis perpendicular to the slot and the z-axis along the rotation axis $(\vec{\omega} = \omega\hat{k})$. A small block of mass m is gently placed in the slot at $\vec{r} = (R/2)\hat{i}$ at t = 0 and is constrained to move only along the slot. **[IIT-JEE Advanced-2016]**



12. The distance r of the block at time t is :

(A) $\frac{R}{4}(e^{2\omega t} + e^{-2\omega t})$ (B) $\frac{R}{2}\cos 2\omega t$ (C) $\frac{R}{2}\cos \omega t$ (D) $\frac{R}{4}(e^{\omega t} + e^{-\omega t})$

13. The net reaction of the disc on the block is :

(A) $-m\omega^2 R\cos\omega t\hat{j} - mg\hat{k}$ (B) $m\omega^2 R\sin\omega t\hat{j} - mg\hat{k}$

(C) $\frac{1}{2}m\omega^2 R(e^{\omega t} - e^{-\omega t})\hat{j} + mg\hat{k}$ (D) $\frac{1}{2}m\omega^2 R(e^{2\omega t} - e^{-2\omega t})\hat{j} + mg\hat{k}$

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Paragraph for question nos. 14 to 16

The potential energy at a point, relative to the reference point is defined as the negative of work done by the conservative force as the object moves from the reference point to the point considered. The value of potential energy at the reference point itself can be set equal to zero because we are always concerned only with differences of potential energy between two points and the associated change of kinetic energy. A particle A is fixed at origin of a fixed coordinate system. Another particle B which is free to move experiences an force $\vec{F}=\left(-\frac{2\alpha}{r^3}+\frac{\beta}{r^2}\right)\hat{r}$ due to particle A where $\vec{r}$ is the position vector of particle B relative to A. It is given that the force is conservative in nature and potential energy at infinity is zero. If B has to be removed from the influence of A, energy has to be supplied for such a process. The ionization energy E_0 is work that has to be done by an external agent to move the particle from a distance r_0 to infinity slowly. Here r_0 is the equilibrium position of the particle.

14. What is potential energy function of particle as function of r.

(A) $\frac{\alpha}{r^2}-\frac{\beta}{r}$ (B) $-\frac{\alpha}{r^2}+\frac{\beta}{r}$ (C) $-\frac{\alpha}{r^2}-\frac{\beta}{r}$ (D) $\frac{\alpha}{r^2}+\frac{\beta}{r}$

15. Find the ionization energy E_0 of the particle B.

(A) $\frac{\beta^2}{2\alpha}$ (B) $\frac{2\beta^2}{\alpha}$ (C) $\frac{\beta^2}{4\alpha}$ (D) $\frac{\beta^2}{\alpha}$

16. If particle B is transfered slowly from point $P_1\left(\sqrt{2}\,r_0, \sqrt{2}\,r_0\right)$ to point $P_2\left(\frac{r_0}{\sqrt{2}}, \frac{r_0}{\sqrt{2}}\right)$ in the xy-plane by an external agent, calculate work required to be done by it in the process.

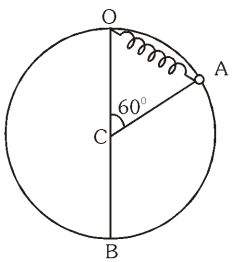
(A) $\frac{9\beta^2}{64\alpha}$ (B) $\frac{\beta^2}{16\alpha}$ (C) $\frac{\beta^2}{64\alpha}$ (D) None of these

SECTION-II

Numerical Answer Type Question **3 Q. [3(0)]**

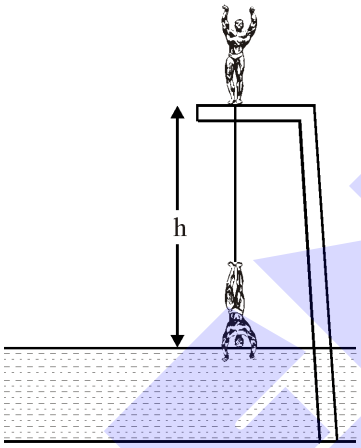
(upto second decimal place)

17. A particle of mass 5 kg is free to slide on a smooth ring of radius r = 20 cm fixed in a vertical plane. The particle is attached to one end of a spring whose other end is fixed to the top point O of the ring. Initially the particle is at rest at a point A of the ring such that $\angle OCA = 60^0$, C being the centre of the ring. The natural length of the spring is also equal to r = 20 cm. After the particle is released and slides down the ring the contact force between the particle & the ring becomes zero when it reaches the lowest position B. Determine the force constant of the spring.



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18. A man of height $h_0 = 2m$ is bungee jumping from a platform situated at a height $h = 25m$ above a lake. One end of an elastic rope is attached to his foot and the other end is fixed to the platform. He starts falling from rest in vertical position. The length and elastic properties of the rope are chosen so that his speed will have been reduced to zero at instant when his head reaches the surface of water. Ultimately the jumper is hanging from the rope with his head 8m above the water. Find the maximum acceleration acheived during the jump in m/s^2 (take $g = 10\ m/s^2$)



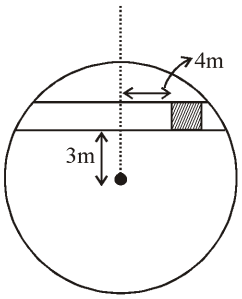
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19. A particle is moving in a circle of radius R in such a way that at any instant the normal and the tangential component of its acceleration are equal. If its speed at t=0 is v_0 then the time it takes to complete the first revolution is $\frac{R}{\alpha v_0}(1-e^{-\beta\pi})$. Find the value of $(\alpha+\beta)^6$.

SECTION-III

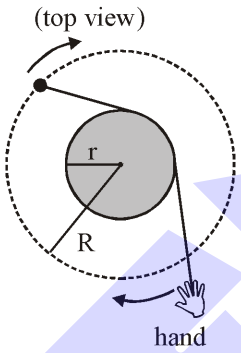
Numerical Grid Type (Ranging from 0 to 9) **8 Q. [4 M (0)]**

20. A small cubical block of mass 1 kg is placed inside a rough rectangular groove made in a circular rough table as shown in the figure. Coefficient of friction for all the rough surfaces is $\mu = 0.5$. The table starts rotating clockwise with angular acceleration 1 rad/sec^2 in a horizontal plane about its axis. Find the time (in sec) after which the block will start motion with respect to table. Assume the size of block slightly smaller then the width of groove.

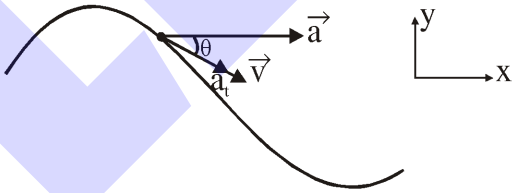


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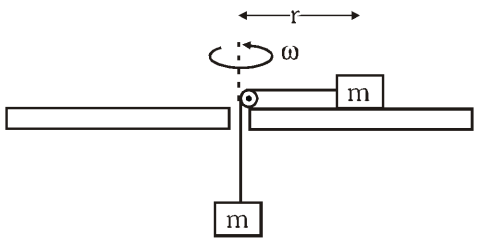
21. A mass m = 1kg, which is free to move on a horizontal frictionless surface, is attached to one end of a massless string that wraps partially around a frictionless vertical pole of radius r (see the top view in figure). You hold on to the other end of the string. At t = 0, the mass has speed v_0 in the tangential direction along the dotted circle of radius R shown. Your task is to pull on the string so that the mass keeps moving along the circular path of radius R. If at any instant 't' force applied by hand on string is f_0 = 2N then find out centripetal acceleration (in m/s^2) of mass. (Given : R = $\frac{2r}{\sqrt{3}}$)



22. A particle moves with a tangential acceleration $a_t = \vec{a}.\hat{v}$ where $\vec{a} = (5\hat{i})$ m/s^2. If the speed of the particle is zero at x = 0, then find v (in m/s) at x = 4.9 m.



23. Two equal masses m are attached by a string. One mass lies at radial distance r from the center of a horizontal turntable which rotates with constant angular velocity ω, while the second hangs from the strings inside the turntable's hollow spindle (see figure). The coefficient of static friction between the turntable and the mass lying on it is μ_s = 0.5. The maximum and minimum values r_{max}, r_{min} of r such that the mass lying on the turntable does not slide. Find the ratio of r_{max} / r_{min} ?



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24. There are three blocks of masses $m_1 = m$, $m_2 = 4m$ and $m_3 = m$. If two block system is kept over a smooth hemisphere with two different arrangement one by one as shown in figure (i) & (ii) respectively are connected with massless & inextensible string. In arrangement figure (i), after releasing the system from rest acceleration of blocks is found to be $\left(a = \frac{5\sqrt{3}}{2}\right) m/s^2$. In second arrangement block m_1 & m_2 are kept symmetrically. Find the acceleration of block (in m/s^2) after releasing the system from rest according to figure (ii).

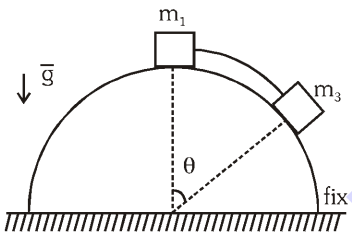


fig. (i)

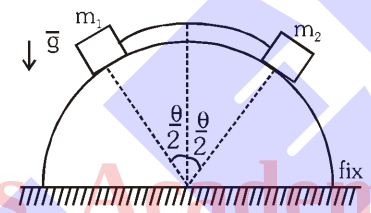
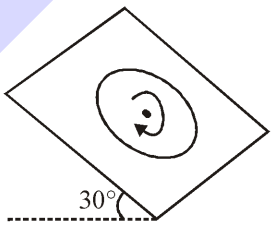


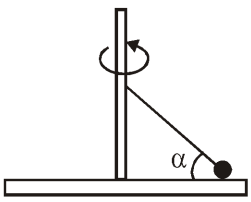
fig. (ii)

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25. An old record player of 10 cm radius turns at 10 rad/s while mounted on a 30° incline as shown in the figure. A particle of mass m can be placed any where on the rotating record. If the least possible coefficient of friction μ that must exist for no slipping to occur is μ, find $2\sqrt{3}\,\mu$.

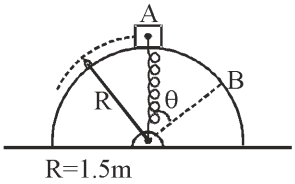


26. A circular platform rotates around a vertical axis with angular velocity $\omega = 10$ rad/s. On the platform is a ball of mass 1 kg, attached to the long axis of the platform by a thin rod of length 10 cm ($\alpha = 30°$). Find normal force exerted by the ball on the platform (in newton). Friction is absent.



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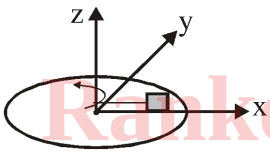
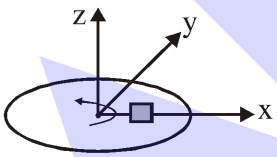
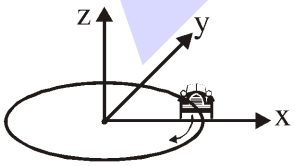
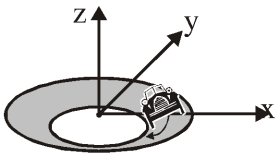
27. A 2 kg block is gently pushed from rest at A and it slides down along the fixed smooth circular surface as shown in figure. If the attached spring has a force constant k = 20 N/m. What is unstretched length of spring (in m) so that it does not allow the block to leave the surface until angle with the vertical is θ = 60°.



SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

28. Mark the matching options for situations shown in respective figures of column-I at the instant, when particle / car is located on the x-axis as shown :

	Column I		**Column II**
(A)	Block attached to string is moving along a circle on rough surface 	(P)	Force due to friction may have non-zero x-component
(B)	Block is placed on a disc rotating with non-uniform angular velocity as shown below. There is no slipping between block & disc. 	(Q)	Force due to friction may have non-zero y-component.
(C)	Car moving on ground along a circular horizontal track at a constant speed 	(R)	Force due to friction may have non-zero z-component.
(D)	Car moving on ground at constant speed along a circular banked track. 	(S) (T)	Force due to friction may be zero. Centripetal force is only due to friction.

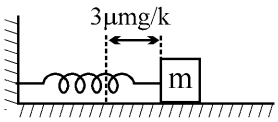
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Subjective Type

2 Q. [4 M (0)]

29. A spring block system is placed on a rough horizontal surface having coefficient of friction μ. Spring is given initial elongation $3\mu mg/k$ (where m = mass of block and k = spring constant) and the block is released from rest. For the subsequent motion find

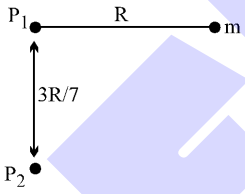


(a) initial acceleration of block

(b) maximum compression in spring

(c) maximum speed of the block

30. A simple pendulum consists of a bob of mass m and a string of length R suspended from a peg P_1 on the wall. A second peg P_2 is fixed vertically below the first one at a distance $3R/7$ from it. The pendulum is drawn aside such that the string is horizontal and released. Calculate the maximum height (with respect to the lowest point) to which it rises



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GR # CIRCULAR MOTION + WORK, POWER & ENERGY — ANSWER KEY

SECTION-I

Single Correct Answer Type — 8 Q. [3 M (–1)]

1. (D) 2. (A) 3. (B) 4. (D) 5. (C)
6. (C) 7. (B) 8. (D)

Multiple Correct Answer Type — 3 Q. [4 M (–1)]

9. (A, B, C) 10. (A,B,C) 11. (ACD)

Linked Comprehension Type (1 Para × 2Q, 1 Para × 3 Q.) [3 M(-1)]
(Single Correct Answer Type)

12. (D) 13. (C) 14. (B) 15. (C) 16. (B)

SECTION-II

Numerical Answer Type Question — 3 Q. [3(0)]
(upto second decimal place)

17. 500 N/m 18. 40 19. 729

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) — 8 Q. [4 M (0)]

20. 2 21. 1 22. 7 23. 3 24. 3
25. 6 26. 5 27. 1

SECTION-IV

Matrix Match Type (4 × 5) — 1 Q. [8 M (for each entry +2(0)]

28. (A) Q, (B) P,Q,T (C) P,T (D) P,R,S

Subjective Type — 2 Q. [4 M (0)]

29. (a) 2μg, (b) μmg/k, (c) $2\mu g\sqrt{\frac{m}{k}}$ 30. 27R/28

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